

Primeval Hunt — Dual Monitor Setup

Sega Lindbergh on Ubuntu 22.04 with NVIDIA GT1030

Technical documentation of the dual monitor configuration process including all attempted approaches, failures, and the final working solution.

Hardware: NVIDIA GeForce GT1030 — Driver 580.159.03

OS: Ubuntu 22.04.5 LTS — GNOME Shell — X11 session

Loader: lindbergh-loader v2.1.4

Monitors: LG TV 32" (HDMI-0) + NCR ATMI15/24SB touchscreen (DVI-D-0)

Game: Primeval Hunt (DVP-0048A) — PRIMEVAL_HUNT_SCREEN_MODE = 2

1. Overview and Goal

Primeval Hunt is a dual-monitor arcade game running on the Sega Lindbergh platform. It uses one monitor for the main gameplay and a second touchscreen monitor for the map, character selection and interactive elements. The lindbergh-loader emulator supports this via **PRIMEVAL_HUNT_SCREEN_MODE = 2** (side-by-side mode), which splits a 1280x480 window into two 640x480 halves — left half for the touchscreen/map, right half for gameplay.

The goal was to configure Ubuntu 22.04 with GNOME/X11 and NVIDIA proprietary drivers to present both monitors as a single 1280x480 logical display to SDL3 (which the loader uses), so that the loader's fullscreen window correctly spans both physical monitors.

Framebuffer Layout (final working configuration):

Left half (+0+0): DVI-D-0 — NCR ATMI15/24SB 15" — touchscreen/map interface

Right half (+640+0): HDMI-0 — LG TV 32" — main gameplay screen

DVI-D-0 always receives the left half (touchscreen/map) and HDMI-0 the right half (gameplay). The physical position of the monitors in the cabinet is independent — in the original Sega cabinet they are stacked vertically.

Primary/Secondary monitor role: The touchscreen monitor (NCR/DVI-D-0) must be configured as the PRIMARY monitor in GNOME display settings, and the gameplay monitor (LG/HDMI-0) as SECONDARY. NVIDIA always places the primary monitor first in the TwinView framebuffer at +0+0 — which is the left half that the game uses for the touchscreen/map interface.

2. Hardware and EDID Analysis

Before attempting any configuration, the EDID (Extended Display Identification Data) of both monitors was extracted and analyzed to understand their supported resolutions and frequency ranges.

2.1 EDID Extraction

NVIDIA proprietary drivers (version 580+) do not expose EDID data via `/sys/class/drm/` as open-source drivers do. Two methods were tried:

Method 1: `get-edid` via DDC/CI (`read-edid` package) — failed, NVIDIA does not expose I2C buses in `/sys/class/drm/`

Method 2: Python parser extracting hex EDID blocks from `xrandr --verbose` output — successful

A custom diagnostic script (`arcade_display_info.sh`) was developed with an interactive menu to choose the extraction method per output, producing separate `edid-decode` and `parse-edid` log files plus binary `.bin` files for each monitor.

2.2 Monitor Specifications

DVI-D-0 — NCR ATMI15/24SB (touchscreen)

Manufacturer: NCR | Model: 26112 | Week 40 of 2019

HorizSync: 28–80 kHz | VertRefresh: 56–80 Hz | Max dotclock: 140 MHz

640x480 natively supported: 59.94 Hz, 72.81 Hz, 75.00 Hz (Established Timings)

Preferred timing: 1024x768 @ 60Hz

HDMI-0 — LG TV 32"

Manufacturer: GSM (LG Electronics) | Week 1 of 2014

HorizSync: 30–83 kHz | VertRefresh: 58–62 Hz | Max dotclock: 160 MHz

640x480 natively supported: 59.94 Hz (VIC 1 in CTA-861 extension block)

Preferred timing: 1920x1080 @ 60Hz

Important: The LG TV has a very narrow vertical refresh range (58–62 Hz). Only 640x480@59.94 Hz is safe. The modeline from `parse-edid` for 640x480 uses pixel clock 25.200 MHz (not the VESA standard 25.175 MHz).

3. Lindbergh-Loader Source Code Analysis

3.1 Dual Screen Logic in resolution.c

Primeval Hunt dual screen setup is handled in resolution.c inside the PRIMEVAL_HUNT case. The critical code block that prevented dual screen from activating:

```
const SDL_DisplayMode *displayMode = SDL_GetCurrentDisplayMode(sdlDisplayId[0]);
if (getConfig()->fullscreen) {
    gWidth = displayMode->w;
    gHeight = displayMode->h;
}
if (gWidth <= 1280 && gHeight <= 480)
    break; // exits without configuring dual screen
```

The check **gWidth <= 1280** caused the problem: with a 1280x480 virtual framebuffer, gWidth equals exactly 1280, satisfying the condition and exiting before the dual screen coordinates (phX, phW, phX2, phW2) are set up.

3.2 SDL Display Enumeration in sdlCalls.c

SDL3 uses SDL_GetDisplays() to enumerate displays. With two physical monitors connected, SDL always finds two separate displays and emits the warning:

```
[0000.000] WARN> More than 1 display detected, will use the first one.
```

It then uses sdlDisplayId[0] (the first physical display, always DVI-D-0 with NVIDIA) to get the display mode for the window. This meant SDL was always working with 640x480 instead of the combined 1280x480 framebuffer.

3.3 Fullscreen Behavior

With FULLSCREEN = true, SDL calls SDL_SetWindowFullscreenMode(window, NULL) followed by SDL_SetWindowFullscreen(window, true). With NULL as the mode, this activates desktop fullscreen on whichever physical display the window is currently on — which was always DVI-D-0 at 640x480, not the combined 1280x480 framebuffer.

4. Attempted Approaches and Why They Failed

4.1 xrandr --fb 1280x480 (virtual framebuffer)

```
xrandr --fb 1280x480 --output HDMI-0 --mode 640x480 --pos 0x0 --output DVI-D-0 --mode 640x480 --pos 640x0
```

Result: SDL3 uses XRandR to enumerate displays and still sees two separate 640x480 displays. The --fb flag creates a virtual framebuffer at the X11 level but SDL3 queries RandR outputs individually, bypassing the virtual framebuffer concept.

4.2 SDL_VIDEO_X11_XRANDR=0 alone

```
SDL_VIDEO_X11_XRANDR=0 ./lindbergh
```

Result: SDL now sees a single display but without the resolution.c patch, gWidth was still 640 from the display mode query so the dual screen check still failed. Additionally, FULLSCREEN=true caused SDL to fullscreen on the first physical display at 640x480.

4.3 Patches to sdlCalls.c — XMoveResizeWindow before SDL_ShowWindow

```
XMoveResizeWindow(x11Display, x11Window, 0, 0, gWidth, gHeight);  
XFlush(x11Display);
```

Result: The window manager (Mutter/GNOME) ignores move/resize requests on unmapped (hidden) windows. The window must be visible before Xlib geometry changes take effect.

4.4 Patches to sdlCalls.c — XMoveResizeWindow after SDL_ShowWindow

```
SDL_ShowWindow(sdlWindow);  
XMoveResizeWindow(x11Display, x11Window, 0, 0, gWidth, gHeight);
```

Result: Debug output showed window moved to x=640,y=0 with size 640x480. Mutter accepted the move but rejected the resize beyond the physical display boundary.

4.5 EWMH _NET_WM_STATE_FULLSCREEN with _NET_WM_FULLSCREEN_MONITORS

```
XSendEvent(..., _NET_WM_FULLSCREEN_MONITORS, top=0, bottom=1, left=0, right=1);  
XSendEvent(..., _NET_WM_STATE, _NET_WM_STATE_ADD, _NET_WM_STATE_FULLSCREEN);
```

Result: Mutter on Ubuntu 22.04 ignores _NET_WM_FULLSCREEN_MONITORS. Window remained at 640x480 on DVI-D-0.

4.6 NVIDIA TwinView alone

```
Option "TwinView" "true"  
Option "MetaModes" "HDMI-0: 640x480 +0+0, DVI-D-0: 640x480 +640+0"
```

Result: TwinView created a single X11 Screen of 1280x480 but SDL3 still uses XRandR to enumerate outputs and found two separate RandR outputs. Additionally NVIDIA ignored MetaModes resolution and defaulted to 720x480.

4.7 TwinView + SDL_VIDEO_X11_XRANDR=0 + FULLSCREEN=true

```
SDL_VIDEO_X11_XRANDR=0 ./lindbergh # with FULLSCREEN=true, WIDTH=1280, HEIGHT=480
```

Result: SDL saw a single display but FULLSCREEN=true called SDL_SetWindowFullscreen which went fullscreen on the first physical display (640x480). Game appeared small on DVI-D-0.

5. Final Working Solution

5.1 Complete Approach

The working solution combines: a patch to resolution.c, a patch to sdlCalls.c, an xorg.conf with NVIDIA TwinView, and the SDL_VIDEO_X11_XRANDR=0 environment variable.

5.2 Patch 1: resolution.c

```
// ORIGINAL:
if (gWidth <= 1280 && gHeight <= 480)
    break;

// PATCHED:
if (gWidth < 1280 && gHeight <= 480)
    break;
```

With TwinView active, gWidth=1280 and gHeight=480. The patched check 1280 < 1280 is false, so execution continues into the dual screen coordinate setup.

5.3 Patch 2: sdlCalls.c

Add include:

```
#include <X11/Xatom.h>
```

Add `SDL_WINDOW_BORDERLESS` to window flags for Primeval Hunt:

```
if (gId == PRIMEVAL_HUNT && getConfig()->fullscreen)
    windowFlags |= SDL_WINDOW_BORDERLESS;
```

Skip `SDL fullscreen` for Primeval Hunt; use `Xlib` after `SDL_ShowWindow`:

```
if (gId == PRIMEVAL_HUNT && getConfig()->fullscreen && x11Display && x11Window)
{
    XMoveResizeWindow(x11Display, x11Window, 0, 0, gWidth, gHeight);
    XRaiseWindow(x11Display, x11Window);
    XFlush(x11Display);
    usleep(100000);
    XMoveResizeWindow(x11Display, x11Window, 0, 0, gWidth, gHeight);
    XFlush(x11Display);
}
```

With TwinView presenting a single 1280x480 X11 Screen, Mutter does not constrain the window to a physical display boundary. XMoveResizeWindow successfully covers the entire framebuffer.

5.4 /etc/X11/xorg.conf — NVIDIA TwinView

```
Option "TwinView" "true"
Option "TwinViewXineramaInfo" "false"
Option "MetaModes" "DVI-D-0: 640x480 +0+0, HDMI-0: 640x480 +640+0"
```

Note: NVIDIA ignores MetaModes resolution at startup and defaults to 720x480. 640x480 must be forced via xrandr after every login.

5.5 lindbergh.ini Settings

```
WIDTH = 1280
HEIGHT = 480
FULLSCREEN = true
PRIMEVAL_HUNT_SCREEN_MODE = 2
```

5.6 Launch Command

```
SDL_VIDEO_X11_XRANDR=0 ./lindbergh
```

Result: No warning, RESOLUTION: 1280x480, dual screen mode 2 activated, touchscreen/map on DVI-D-0 (left half), gameplay on HDMI-0 (right half).

6. Replication on Another Machine

This section explains how to adapt the configuration for a different machine with different monitors. The EDID data is the critical variable — everything that depends on the physical monitors must be verified before proceeding.

6.1 Why EDID Matters

EDID (Extended Display Identification Data) is the information stored in each monitor that tells the graphics driver what resolutions, refresh rates and timings it supports. The following configuration parameters depend entirely on the EDID of the monitors used:

- **Output names** (HDMI-0, DVI-D-0, DP-0, etc.) — may differ on another machine
- **640x480 availability** — not all monitors declare it; may need a custom modeline
- **Pixel clock for 640x480** — VESA standard is 25.175 MHz but some monitors require a different value (e.g. LG TV required 25.200 MHz)
- **HorizSync and VertRefresh ranges** — the modeline must fall within these ranges
- **MetaModes resolution in xorg.conf** — must use a resolution both monitors support
- **xrandr --rate value** — must match a rate the monitor actually supports

What does NOT change between machines: the patches to resolution.c and sdlCalls.c, PRIMEVAL_HUNT_SCREEN_MODE = 2, SDL_VIDEO_X11_XRANDR=0, and the TwinView approach itself.

6.2 Step 1 — Run arcade_display_info.sh

The first thing to do on any new machine is run the diagnostic script **arcade_display_info.sh** with both monitors connected and the X11 session active. This script collects all the EDID and display information needed to adapt the configuration.

```
./arcade_display_info.sh
```

The script produces several files in the same directory:

- **arcade_display_info_TIMESTAMP.log** — main log with xrandr output, frequency ranges, modeline test results
- **edid-decode_OUTPUT_TIMESTAMP.log** — full EDID decode for each monitor (one file per monitor)
- **parse-edid_OUTPUT_TIMESTAMP.log** — EDID in xorg.conf format with HorizSync, VertRefresh and modelines ready to use
- **edid_OUTPUT_TIMESTAMP.bin** — raw binary EDID for archiving

When prompted, choose Method 1 (get-edid via DDC/CI) first. If it fails (common with NVIDIA proprietary drivers), the script will offer Method 2 (xrandr --verbose parser) automatically.

6.3 Step 2 — Read the Results

From the generated files, extract the following information:

From arcade_display_info_TIMESTAMP.log, section 2 (xrandr --query):

Note the exact output names (e.g. HDMI-0, DVI-D-0, DP-0) and which monitor is connected to each output. These names will be used in xorg.conf MetaModes and in the xrandr commands.

Also check section 5 (640x480 availability). If you see:

- ✓ 640x480 PRESENT — use the native mode directly with xrandr --mode 640x480
- ✗ 640x480 NOT present — you need to create a custom modeline (see step 3)

From parse-edid_OUTPUT_TIMESTAMP.log:

Look for:

- **Horizsync** and **VertRefresh** values — copy these into the Monitor sections of xorg.conf
- **Modeline entries** — if 640x480 is not in the EDID, use the modeline parameters shown here to create a compatible custom modeline. The pixel clock (first number after the mode name) is critical — it must match what the monitor expects.

From edid-decode_OUTPUT_TIMESTAMP.log:

Verify under 'Established Timings I & II' whether 640x480 appears. Also check 'Display Range Limits' for the exact HorizSync and VertRefresh ranges. Any custom modeline you create must have a pixel clock and sync frequencies that fall within these declared ranges, otherwise the monitor will reject the signal.

6.4 Step 3 — Adapt xorg.conf

Using the information gathered, modify the following in xorg.conf:

BusID — get the correct value with:

nvidia-xconfig --query-gpu-info | grep PCI

```
nvidia-xconfig --query-gpu-info | grep PCI
```

Output names in MetaModes — replace DVI-D-0 and HDMI-0 with the actual output names from xrandr --query. The output that must be on the LEFT (position +0+0) is the one connected to the touchscreen/map monitor. DVI-D-0 receives the left half (touchscreen/map) and HDMI-0 the right half (gameplay). Physical monitor placement in the cabinet is independent.

HorizSync and VertRefresh in Monitor sections — use the values from parse-edid.

If 640x480 is not in the EDID — add a Modeline to the Monitor section. The modeline must use a pixel clock compatible with the monitor's declared ranges. Use cvt or gtf to generate a starting point, then verify against the EDID ranges:

```
cvt 640 480 60 # generates a modeline for 640x480 @ 60Hz
```

Then add to the Monitor section in xorg.conf:

```
Modeline "640x480" [paste cvt output here]
```

6.5 Step 4 — Adapt the xrandr Commands

NVIDIA with TwinView ignores MetaModes at startup and defaults to a different resolution. After every login, force 640x480 with xrandr using the actual output names and a rate supported by each monitor:

```
xrandr --output [LEFT_OUTPUT] --mode 640x480 --rate [RATE] --pos 0x0
xrandr --output [RIGHT_OUTPUT] --mode 640x480 --rate [RATE] --pos 640x0
```

If 640x480 is not natively available on a monitor, add the custom modeline first:

```
xrandr --newmode '640x480_custom' [modeline parameters]
xrandr --addmode [OUTPUT] 640x480_custom
xrandr --output [OUTPUT] --mode 640x480_custom --pos [position]
```

Use the rate value from xrandr --query output (the number shown next to 640x480 in the monitor's mode list, e.g. 59.94 or 75.00). For monitors where 640x480 is not listed, use the refresh rate of the custom modeline you created.

6.6 Step 5 — Verify and Launch

After applying xorg.conf and running the xrandr commands, verify:

```
xrandr --query
```

Expected output:

```
Screen 0: current 1280 x 480 ← TwinView single framebuffer  
[LEFT_OUTPUT] connected 640x480+0+0 ← touchscreen/map monitor  
[RIGHT_OUTPUT] connected 640x480+640+0 ← touchscreen monitor
```

Then launch the game:

```
SDL_VIDEO_X11_XRANDR=0 ./lindbergh
```

The log should show:

```
RESOLUTION: 1280x480  
(no 'More than 1 display detected' warning)
```

If RESOLUTION shows 640x480 instead of 1280x480, the TwinView framebuffer is not being seen correctly. Verify that xrandr --query shows 'current 1280 x 480' and that SDL_VIDEO_X11_XRANDR=0 is set.

7. Provided Files and Scripts

Loader patches (recompile required):

- **resolution.c** — patched source file for lindbergh-loader v2.1.4
- **sdlCalls.c** — patched source file for lindbergh-loader v2.1.4

Copy both to src/lindbergh/ and run: make clean && make

System configuration:

- **xorg.conf** — NVIDIA TwinView configuration for this specific hardware. Adapt BusID, output names and HorizSync/VertRefresh for different hardware.

Copy to /etc/X11/xorg.conf and restart display manager.

Scripts:

- **arcade_display_info.sh** — diagnostic script to extract and decode EDID from both monitors. Run this first on any new machine.
- **launch_primeval_hunt.sh** — sets monitors to 640x480 via xrandr and launches the game with SDL_VIDEO_X11_XRANDR=0. Run after every reboot before playing.

Known remaining issues:

- GNOME top bar and dock overlay the game. Fix: install Just Perfection GNOME extension and run: gsettings set org.gnome.shell.extensions.just-perfection panel false / dash false
- NVIDIA TwinView ignores MetaModes resolution at startup — xrandr reset required after each login (handled by launch_primeval_hunt.sh)
- SDL_VIDEO_X11_XRANDR=0 must be set for every game launch (handled by launch_primeval_hunt.sh)